Prompt Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Prompt Experiments

> Prepare Prompt Experiments: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of controlled **prompt** experiments

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/28, ai-3/7, ai-3/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

Text as Pieces

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and decode text under an explicit toy vocabulary

Reference example, not the only acceptable wording: Teacher explanation: The rule chooses the longest matching ordinary token; sat has length three and s length one.

Vocabulary and matching are case sensitive and contain no uppercase C.

Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank **training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

Concept to explain

Prompt quality is evaluated by **controlled comparison**s and success criteria.

Term	Meaning
prompt	Input instructions and context supplied to a generative system.
constraint	A condition that a valid solution must satisfy.
criterion	One stated standard used to judge a result.

Term	Meaning
controlled comparison	A comparison designed to keep relevant conditions alike except for the factor being studied.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A **prompt** experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can compare **prompt**s on **paired case**s and inspect a subgroup before choosing.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Prompt Experiments using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Prompt** A meets 3 of 6 **rubric** checks.

2. Add one **constraint**: ‘Use 3 sentences and name no ending.’

3. **Prompt** B meets 5 of 6 checks.

4. Improvement = +2 checks; repeat on more books before claiming it generalizes.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

The teaching case

Design a fair comparison.

Exact explanation to model

Why the result follows

Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

Explain the added worked example

A **prompt** experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Evaluate both **prompt**s on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Paired case	The same case evaluated under each compared procedure
Rubric	Prespecified criteria for judging an output
Regression	A previously passing condition that fails after a change

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Prompt Experiments

A fictional book-help project tests two **prompt** versions on seven shared development cases.

The facts we will use

> **Supplied facts**
>
> P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Supplied facts

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

AI ACADEMY / AI-2/28

Paired overall and missing-source passes

	Overall passes /7	Missing-source passes /2
P	4	1
Q	5	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/28

Prompt Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired setup

Use the same seven cases and pass criteria for both prompts.



2. Overall rates

P=4/7; Q=5/7.



3. Missing-source slice

P passes 1/2; Q passes 0/2 in cases 4 and 7.



4. Decide explicitly

Q improves overall count but worsens missing-source behavior; use the declared priorities.

CHANGE ONE THING

Make missing-source correctness a required selection gate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Paired setup	Use the same seven cases and pass criteria for both prompts.
2. Overall rates	P=4/7; Q=5/7.
3. Missing-source slice	P passes 1/2; Q passes 0/2 in cases 4 and 7.
4. Decide explicitly	Q improves overall count but worsens missing-source behavior; use the declared priorities.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

What this example does not establish

Keep the stated limit

Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

Change one thing in the pictured case

Change one thing

Change: Make missing-source correctness a required selection gate.

Predicted result

Prediction: Neither **prompt** clears a 2/2 gate on these records.

Why the result follows

Reason: P fails 7 and Q fails both 4 and 7; both need revision.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Prompt Experiments, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Prompt** B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

2. State the decision to explain

Write the question to explain: Design a fair comparison.

3. Reveal the worked reasoning

Reveal after **prediction**s: Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

Hold back the independent task

Keep for an independent attempt: **Prompt** A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a **controlled comparison**.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Qualification: **Prompt** selection on development cases cannot be reported as independent final performance.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage	Teacher action
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Controlled **prompt** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: **Prompt** selection on development cases cannot be reported as independent final performance.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Name the case P passes and Q fails, and explain why it matters.

Reasoned teacher answer

Case 4, a missing-source case. Its **regression** can matter more than gains on ordinary cases if unsupported answers are unacceptable.

Changed-case reasoning: Neither **prompt** clears a 2/2 gate on these records. P fails 7 and Q fails both 4 and 7; both need revision.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Prompt Experiments investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Define six **rubric** checks, keep task and examples fixed, change one **prompt** element, score both versions across three cases, and explain uncertainty.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: **Prompt** selection on development cases cannot be reported as independent final performance.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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Assess and explain the answers

> Use the Prompt Experiments questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

Independent question

Independent question · withhold the key

Prompt A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a **controlled comparison**.

Expected independent answer

Teacher key · Expected independent answer

No; question difficulty also changed. Compare both **prompt**s on the same evaluation questions and **rubric**, with repeated runs when **output**s vary.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Changing several factors proves which one caused improvement.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> Run both **prompt**s on the same predefined requests and catalogue, with identical criteria for existence, **constraint**s and explanation support. Repeat stochastic runs and retain failures.
>
> Separate **prompt**-development cases from final evaluation and report per-case outcomes, not only the mean.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: A **prompt** experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Evaluate both **prompt**s on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

W2 Trace the worked case

Workbook question · W2

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: **Prompt** selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: **Prompt** selection on development cases cannot be reported as independent final performance.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

Teacher reasoning · W4

Teacher explanation: A=5/8, B=6/8, so select B on development. Special-case passes A=1/2, B=0/2; the aggregate selection hides a **regression** requiring review.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Prompt** selection on development cases cannot be reported as independent final performance.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: A=5/8, B=6/8, so select B on development. Special-case passes A=1/2, B=0/2; the aggregate selection hides a **regression** requiring review.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design **prompt** is a starting brief, not automatically an equivalent test.

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Connect and extend learning

> Connect Prompt Experiments to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Test generated book explanations with a fixed **rubric** and prohibit invented details.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 29: **Evidence** and Sources. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: A=5/8, B=6/8, so select B on development. Special-case passes A=1/2, B=0/2; the aggregate selection hides a **regression** requiring review. Require intermediate reasoning and the stated scope.

Fresh retry: A passes 1–4 of six; B passes 1, 2, 5. Special cases 4 and 6. Compare. Expected reasoning: A=4/6, B=3/6; special A=1/2, B=0/2. Preserve both outcomes; no final result supplied.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Generation failures and verification. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-2/29, ai-2/31, ai-2/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Prompt Experiments lesson; it is a

checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of controlled **prompt** experiments

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Check every learner before discussion: Use the worked case for Controlled **prompt** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Prompt** selection on development cases cannot be reported as independent final performance.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Make missing-source correctness a required selection gate. Require a **prediction** and a reason before comparing. Expected result: Neither **prompt** clears a 2/2 gate on these records.

Reason: P fails 7 and Q fails both 4 and 7; both need revision.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

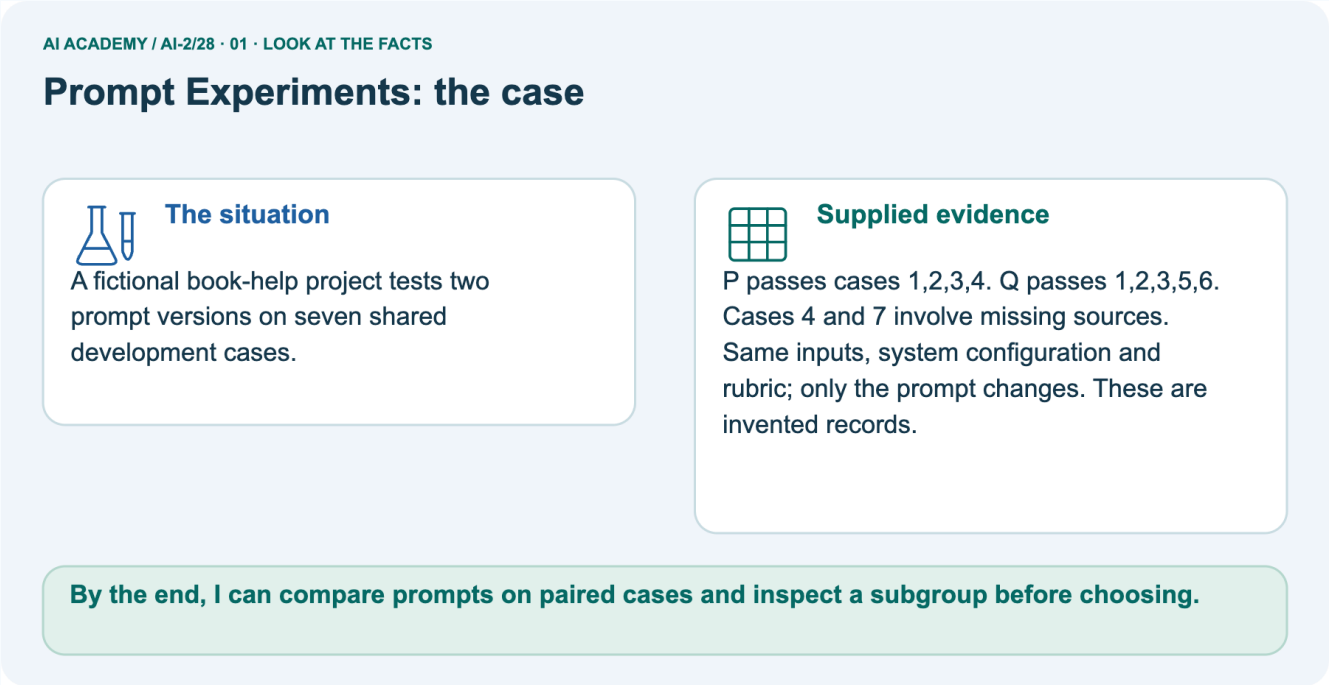
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Visual concept journey

> Show one Prompt Experiments graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Prompt Experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-help project tests two **prompt** versions on seven shared development cases. Supplied facts: P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

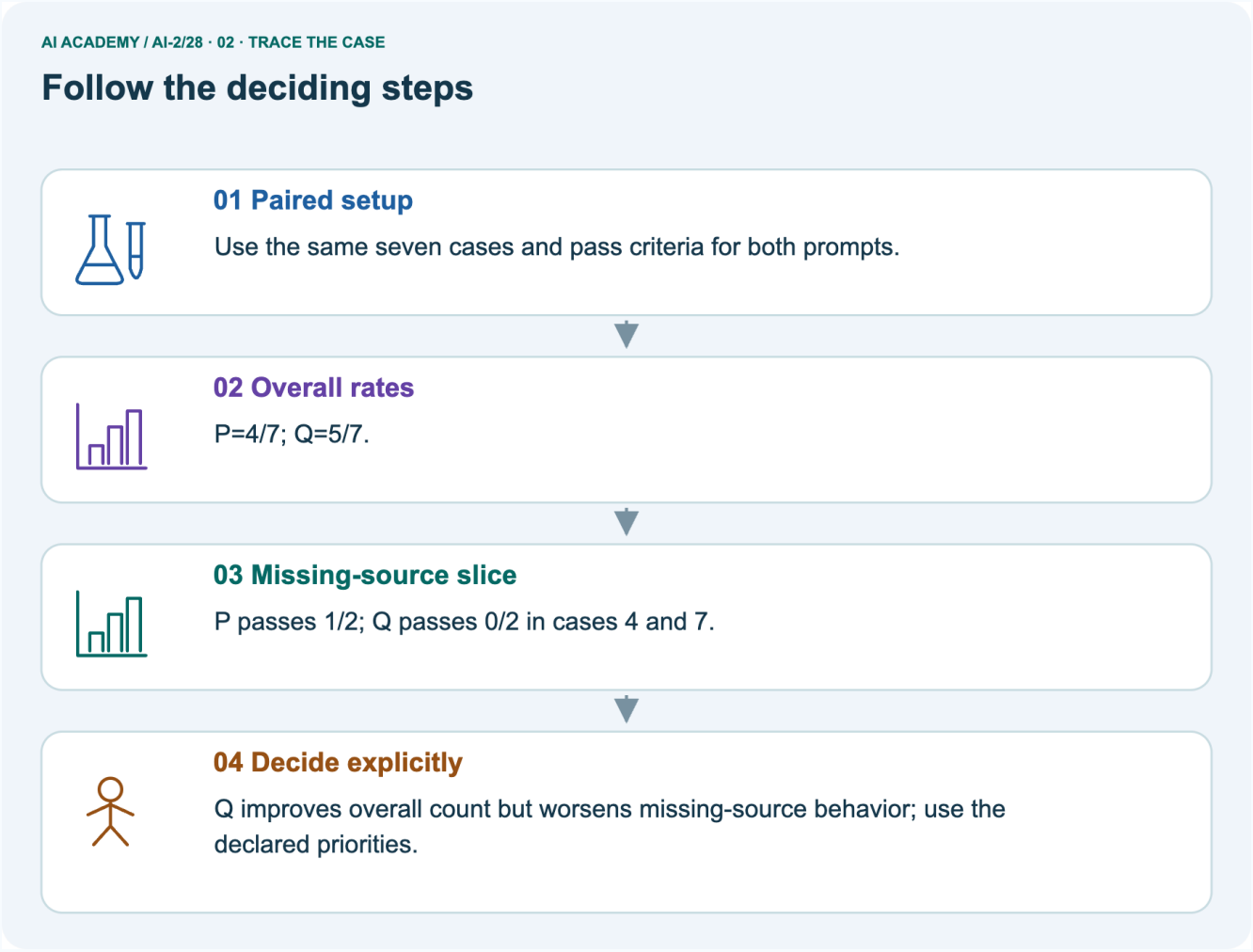
Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher **prompt**: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

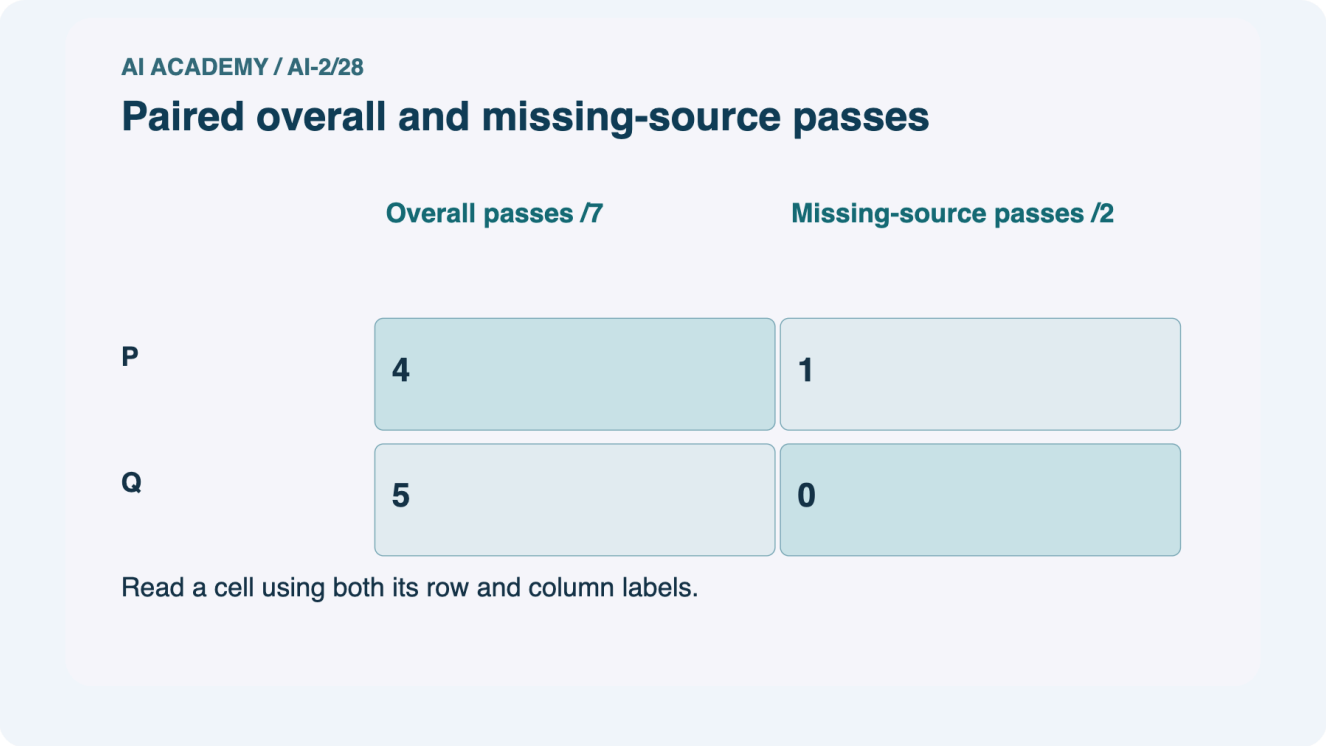
Paired setup: Use the same seven cases and pass criteria for both **prompt**s. → Overall rates: P=4/7; Q=5/7. → Missing-source slice: P passes 1/2; Q passes 0/2 in cases 4 and 7. → Decide explicitly: Q improves overall count but worsens missing-source behavior; use the declared priorities.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher **prompt**: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Paired overall and missing-source passes



Follow the labelled picture with the matching explanation. Open the vector diagram

Paired overall and missing-source passes. P: Overall passes /7 = 4, Missing-source passes /2 = 1. Q: Overall passes /7 = 5, Missing-source passes /2 = 0.

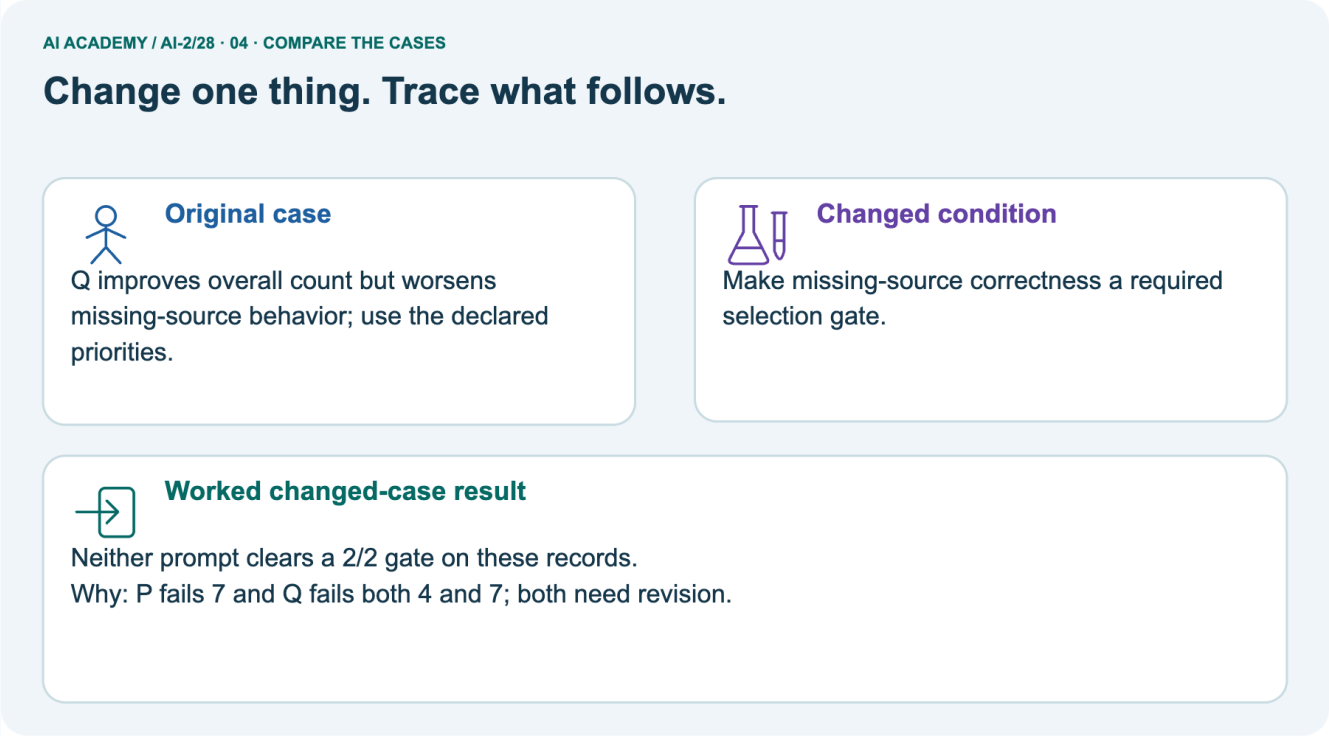
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher **prompt**: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Q improves overall count but worsens missing-source behavior; use the declared priorities. Changed condition: Make missing-source correctness a required selection gate.

Expected result: Neither **prompt** clears a 2/2 gate on these records. Reason: P fails 7 and Q fails both 4 and 7; both need revision.

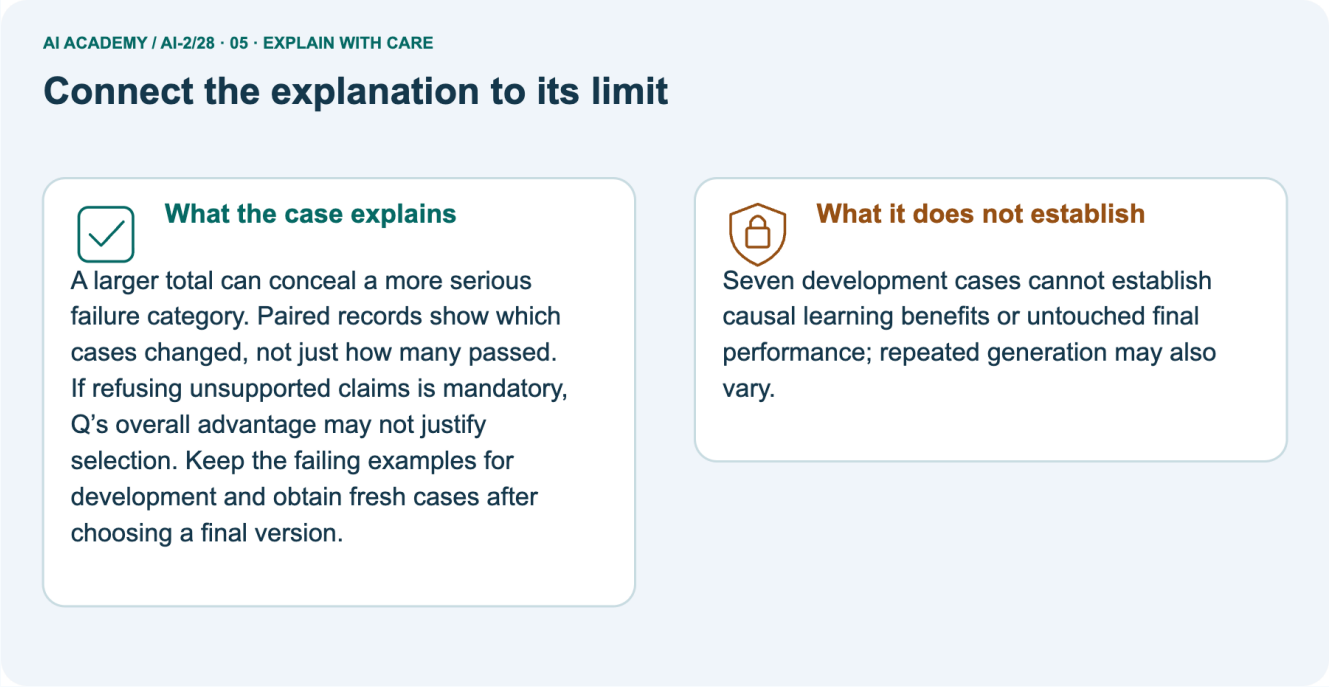
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher **prompt**: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: P fails 7 and Q fails both 4 and 7; both need revision.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Limit: Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

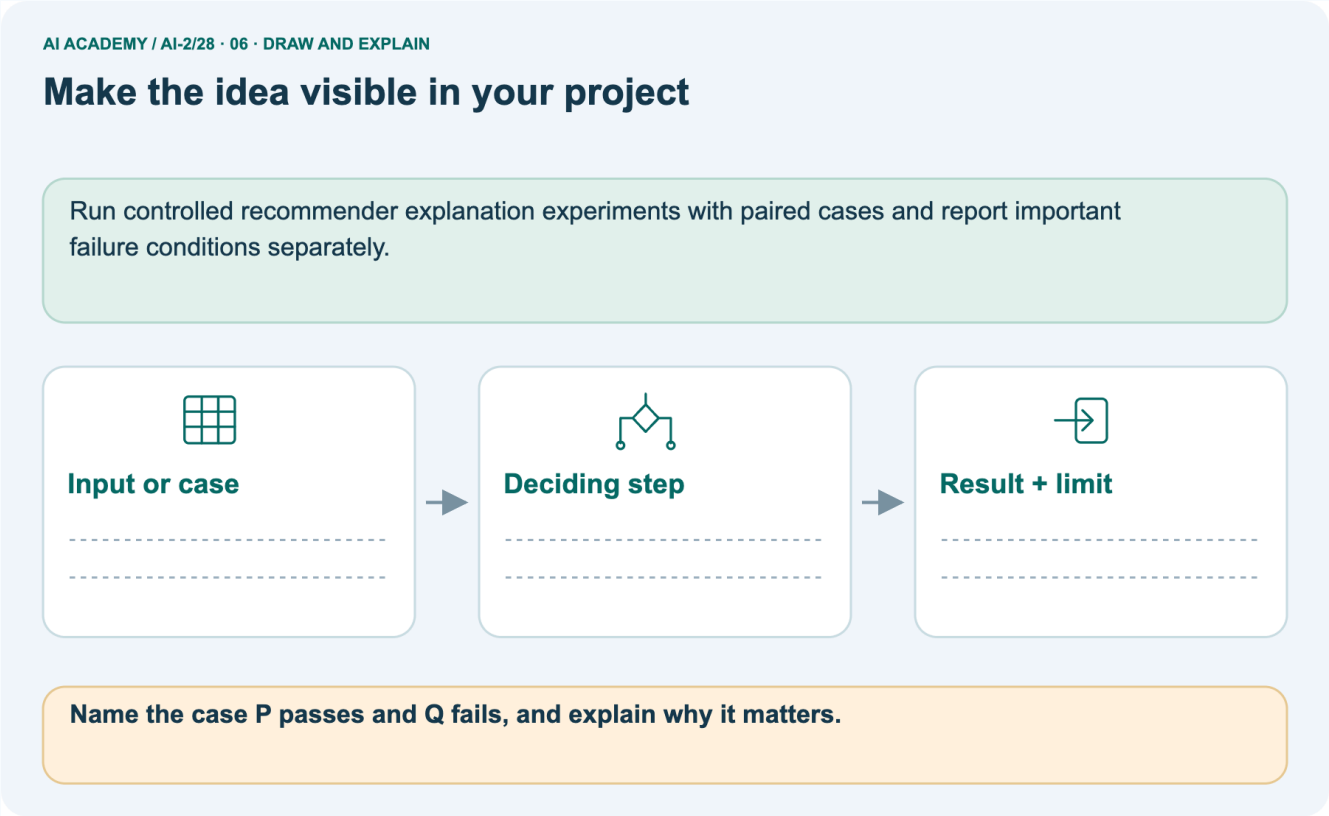
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher **prompt**: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately. Draw your **input**, deciding step, result and limit.

Supported question: Name the case P passes and Q fails, and explain why it matters.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher **prompt**: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Case 4, a missing-source case. Its **regression** can matter more than gains on ordinary cases if unsupported answers are unacceptable.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.